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FACULTY OF MATHEMATICS AND COMPUTER SCIENCE
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DISSERTATION THESIS

Social Media Bots in the Age of Generative AI: Detection, Simulation, and Electoral Influence

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2026

ABSTRACT

Social media bots, automated accounts designed to simulate human presence on digital platforms, have evolved from simple spam scripts into sophisticated synthetic personas powered by generative artificial intelligence. This evolution has destabilised the established methods for detecting them and has, in several documented cases, produced consequences at the level of national elections. The 2024 Romanian presidential election, annulled by the Constitutional Court following evidence of a large-scale coordinated influence operation conducted through TikTok and Telegram, stands as the first case in European history in which a national vote was invalidated on the basis of digital interference, and it serves as the central case study of this work.

This dissertation examines social media bots from a technical standpoint, structured around three connected questions: how modern automated accounts are built and deployed, how machine learning methods can be designed to detect them, and how the arrival of large language models and other generative tools reshapes both sides of that equation. The theoretical chapters survey the state of the art in detection, examine the supporting infrastructure that enables operations at scale, including the SIM farm economy recently exposed by Europol's Operation SIMCARTEL, and analyse a series of real-world influence operations across Eastern Europe, with particular attention to the Romanian case.

The dissertation contributes two experimental studies. The first designs and evaluates a bot detection system trained on public benchmark data, with explicit attention to cross-dataset generalisation and to the methodological pitfalls that have produced inflated accuracy figures in earlier work. The second uses a fine-tuned language model to generate synthetic accounts and tests the resulting personas against the detection system, framing the exercise as defensive red-teaming rather than as an exercise in offence. The work closes with a discussion of platform-level and regulatory countermeasures, including the early implementation of the European Union's Digital Services Act, and with reflections on the institutional gaps that the Romanian case made visible.

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Chapter 1

Introduction

On the evening of November 24, 2024, Romania's Constitutional Court issued a decision with no precedent in European democratic history: it annulled the first round of a presidential election. The justification pointed not to ballot irregularities, but to evidence of a coordinated digital influence campaign conducted through social media, primarily TikTok, in which thousands of automated accounts artificially amplified a single candidate to an audience of millions [Fun25]. The election had been manipulated, and the instrument of that manipulation was software.

This event was not an isolated incident. For over a decade, social media bots, that is, automated accounts designed to simulate human presence on digital platforms, have been documented in operations targeting elections, financial markets, and public health debates worldwide [FVD+16]. What began as unsophisticated spam accounts has evolved, rapidly and quietly, into something far more dangerous: synthetic personas powered by the same generative artificial intelligence that underlies modern language models. They write fluently, adapt to context, and scale to sizes no human team could match. The result is an information environment in which the boundary between organic public opinion and manufactured consensus is increasingly difficult to locate.

This dissertation approaches the problem from a technical standpoint, structured around three interconnected questions: how are modern bots built; how can machine learning detect them; and how does generative AI complicate both sides of that equation? The work makes two original experimental contributions. First, a multi-model bot detection system is designed and evaluated across publicly available benchmark datasets, with preprocessing pipelines built on distributed computing infrastructure appropriate to the scale of real-world social network data. Second, a synthetic bot generation experiment is conducted using a fine-tuned language model, with the generated accounts then used to adversarially stress-test the detection system. These experiments are framed not as exercises in creating harmful tools, but as a form of red-teaming: one cannot build a reliable detector without first understanding what it needs to detect.

The theoretical chapters situate this technical work in context. A systematic review of the state of the art covers bot taxonomies, detection methodologies, and the shift introduced by generative AI. A comparative case study chapter examines documented influence operations across Eastern Europe, with the 2024 Romanian presidential election as the central example, chosen because it is, as of this writing, the most consequential instance of bot-driven electoral interference in European history and the only case in which a democratic election was formally annulled as a direct result [Fun25]. The dissertation concludes with an analysis of counter-strategies and the limitations of current regulatory instruments, including the EU Digital Services Act.

The remainder of this work is organized as follows. Chapter 2 reviews the literature on social media bots and detection methods. Chapter 3 examines real-world influence operations, with extended analysis of the Romanian case. Chapter 4 covers bot creation techniques and infrastructure. Chapter 5 surveys detection methodologies in depth. Chapters 6 and 7 present the two original experiments. Chapter 8 discusses defenses and regulation. Chapter 9 summarizes findings and proposes directions for future work.

Chapter 2

State of the Art

2.1 Defining the Problem: What Is a Social Media Bot?

The term *social media bot* lacks a single agreed-upon definition in the literature, which is itself a symptom of how rapidly the phenomenon has evolved. For the purposes of this dissertation, we adopt the working definition proposed by [FVD+16]: a social media bot is a software-controlled account that operates on a social platform with the intent to simulate, or pass as, a human user. This definition is deliberately broad. It covers simple scripted accounts that repost content at fixed intervals, coordinated networks of accounts that amplify specific narratives in concert, and the newer generation of large-language-model-driven personas that generate original text indistinguishable from human writing.

What unifies these otherwise very different systems is the combination of *automation* and *deception*: the account acts without direct human control in each individual interaction, and it conceals that fact from the platform and from other users.

2.2 A Brief History: Three Generations of Bots

The academic study of social media automation traces its origins to the rise of Twitter in the late 2000s. Early bots were relatively transparent: news aggregators, weather services, and earthquake alert systems that made no serious attempt to appear human. Research interest shifted when it became clear that the same infrastructure could be turned toward manipulation.

[Cre20] identifies three broad generations of bots, summarized in Figure 2.1.

The transition from the second to the third generation is the defining development of the current period. The introduction of transformer-based language models into bot pipelines means that content quality is no longer a reliable discriminator: a generative-AI-powered bot produces grammatically correct, contextually relevant,

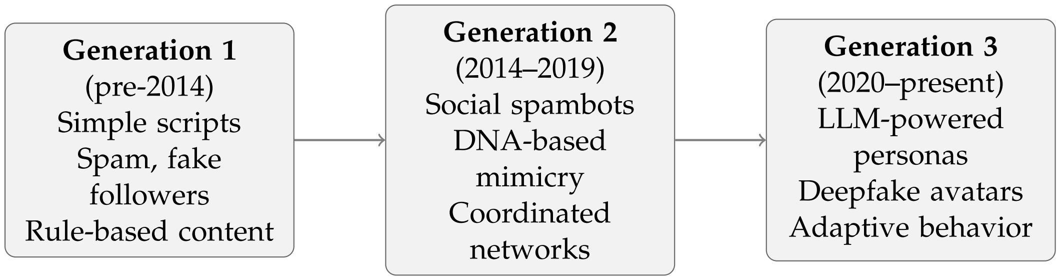

Figure 2.1: Three generations of social media bots, Each generation is substantially harder to detect than its predecessor.

and stylistically varied text by default. The detection community has not yet reached consensus on how to handle this shift [FW+24].

2.3 Scale and Impact

Estimates of bot prevalence vary considerably, partly because platforms are reluctant to disclose internal figures and partly because different methodologies produce different results. A widely cited study found that during the 2016 US presidential election, approximately 400,000 automated accounts produced around 20% of all election-related social media content in the final 24 hours before voting [FVD+16]. More recent academic estimates place the bot population on Twitter/X at between 8 and 18% of all accounts [HSR+23a], though some analyses using proprietary tools report considerably higher figures.

The consequences extend well beyond politics. Bots have been documented manipulating stock prices through coordinated false narratives, amplifying health misinformation during the COVID-19 pandemic, and distorting engagement metrics in ways that determine what content is algorithmically promoted to human users. The fundamental problem is one of signal integrity: when a meaningful fraction of apparent social consensus is artificially manufactured, the information environment itself becomes unreliable.

2.4 Detection Methods

The detection literature organizes broadly into three families of approaches, each with distinct strengths and limitations [AZS+23].

2.4.1 Feature-Based Methods

The earliest and most widely applied detection approaches rely on handcrafted features derived from observable account properties. These features fall into four principal categories, illustrated in Figure 2.2.

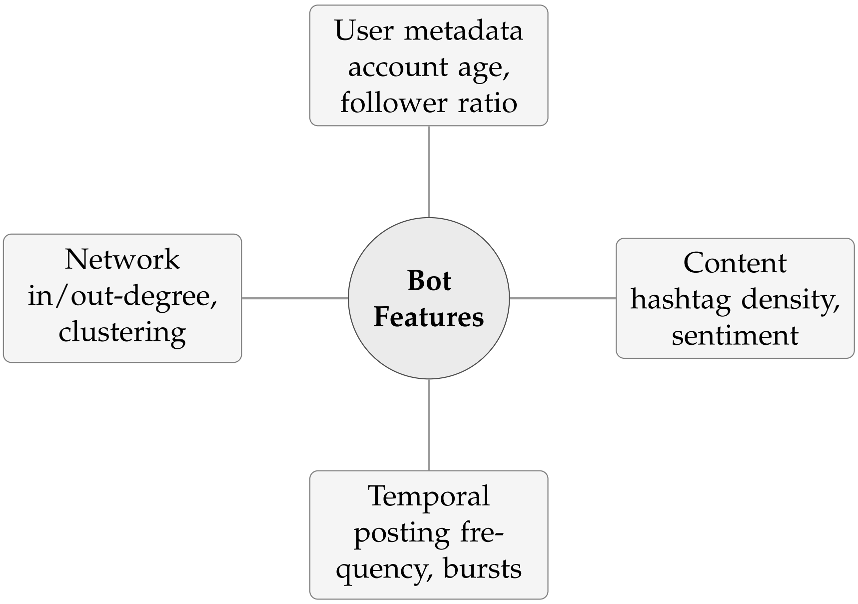

Figure 2.2: The four principal feature categories used in social bot detection. The Botometer system uses over 1,200 features distributed across these categories.

Botometer, developed at Indiana University and described in [VFD+17a], is the most widely deployed feature-based system. It aggregates over 1,200 features into a single probability score and has processed hundreds of millions of account queries. Despite its adoption, Botometer has been shown to fail on newer LLM-driven bots, achieving recall as low as 4% on certain post-2022 datasets [C+24]. This performance collapse is the clearest empirical evidence of the generational problem described in Section 2.2.

Classical machine learning algorithms applied to these features, principally Random Forest, Support Vector Machines, and Gradient Boosting, achieve competitive accuracy on older benchmark datasets. The best-performing models on the Cresci-2017 dataset exceed 95% accuracy [HSR+23a]. Crucially, however, [HSR+23a] demonstrate that this figure is largely an artifact of dataset construction: a simple decision tree using only account creation timestamps achieves comparable results on the same data, because bot and human accounts in the dataset were collected at systematically different times. This finding has important methodological implications for the experimental work in Chapter 6.

2.4.2 Deep Learning and Sequence Models

The second generation of detection methods applies deep learning to the sequential nature of user behavior. Long Short-Term Memory networks model the temporal evolution of posting activity, while Convolutional Neural Networks extract local patterns from content. Fine-tuned transformer models, particularly BERT and RoBERTa, have improved content-based classification by representing tweet semantics in dense vector spaces rather than through handcrafted linguistic features.

The practical challenge with deep learning approaches is generalization. A model trained on data from 2017 has learned to recognize the behavioral signatures of bots that existed in 2017. The bots of 2024, optimized against precisely these detection systems, exhibit different signatures [Cre20]. This phenomenon, known as concept drift, represents one of the central unsolved problems in the field.

2.4.3 Graph Neural Network Methods

The most recent line of research treats bot detection as a node classification problem on the social network graph. Rather than classifying each account in isolation, graph-based methods exploit the structural relationships between accounts: who follows whom, who mentions whom, which accounts retweet the same content within narrow time windows.

[FTLL21b] introduce BotRGCN, a relational graph convolutional network that processes the heterogeneous social graph of Twitter/X and achieves state-of-the-art results on the TwiBot-20 and TwiBot-22 benchmarks. The mathematical formulation of a single graph convolution layer is given in Equation 2.1, where $\mathbf{H}^{(l+1)}$ denotes the node representation matrix at layer $l+1$, $\tilde{\mathbf{A}}$ is the normalized adjacency matrix with self-loops, and $\mathbf{W}^{(l)}$ is the learnable weight matrix:

$$\mathbf{H}^{(l+1)} = \sigma\left(\tilde{\mathbf{A}}\,\mathbf{H}^{(l)}\,\mathbf{W}^{(l)}\right) \tag{2.1}$$

Graph-based methods are particularly effective at detecting coordinated inauthentic behavior, where the network topology itself is anomalous even if individual accounts appear plausible in isolation.

2.5 The Generative AI Shift

The integration of large language models into bot pipelines has introduced a qualitative change in the threat landscape. Earlier bots were detectable, at least in part, because their content was mechanically generated: repetitive, grammatically limited, and stylistically uniform. Current-generation bots, built on fine-tuned versions of models such as GPT or Llama, produce text that is contextually coherent, stylistically varied, and responsive to conversational context [FW+24].

This creates a direct problem for content-based detectors. If a bot generates text that is genuinely indistinguishable from human writing, then any classifier trained to distinguish them based on textual features will perform at chance. Detection must therefore shift toward behavioral and network signals, which are harder to spoof at scale, and toward adversarial training regimes that expose detection systems to the kinds of content they will face in deployment.

The experimental chapters of this dissertation engage directly with this problem. Chapter 6 evaluates whether modern detection models can maintain acceptable performance on recent benchmark data. Chapter 7 tests this boundary explicitly by generating synthetic bot accounts with a fine-tuned language model and measuring evasion rates against the trained detector.

2.6 Benchmark Datasets and Their Limitations

Any experimental contribution in this space must engage with the available benchmark datasets and their well-documented limitations. The most widely used datasets are summarized in Table 2.1.

Table 2.1: Principal benchmark datasets for social bot detection research.

Dataset	Accounts	Tweets	Access	Key limitation
Cresci-2017	8,377	561K	Open	Temporal sampling bias
TwiBot-20	229,573	33.5M	Open	Limited graph data
TwiBot-22	926,949	139M	Request	Requires approval

The critique of benchmark validity raised by [HSR$^+$23a] applies most acutely to Cresci-2017, but is a general concern across the literature. Bot accounts and human accounts are often collected through different pipelines at different times, introducing spurious correlations that naive classifiers exploit without learning anything meaningful about bot behavior. Cross-dataset evaluation, in which a model is trained on one dataset and tested on another, is the standard remedy and is adopted as a core evaluation practice in Chapter 6.

2.7 Regulatory and Platform Responses

The technical detection literature has developed largely in parallel with, and in tension with, platform-level and regulatory responses. Social media platforms have internal trust and safety teams and proprietary detection systems, but they are not required to disclose their methods or their results, which makes independent validation impossible. The EU Digital Services Act, which came into full force in 2024, introduces new transparency and accountability obligations for very large online platforms, including requirements to assess and mitigate systemic risks related to disinformation [Eur24]. Whether these obligations translate into measurable improvements in bot suppression remains an open empirical question; the Romanian case discussed in Chapter 3 provides early and troubling evidence that regulatory instruments have not yet caught up with operational realities.

2.8 Research Gaps

The review above identifies four principal gaps that motivate the contributions of this dissertation. First, most detection benchmarks predate the widespread availability of generative AI tools and do not reflect the capabilities of current bot infrastructure. Second, cross-dataset generalization is rarely evaluated systematically, limiting the practical relevance of reported accuracy figures. Third, the adversarial dimension of detection, specifically, the question of whether a well-designed generative system can evade a well-designed detector, has received limited empirical attention. Fourth, the connection between technical bot detection research and the documented behavior of real-world influence operations is often loose; case studies and detection experiments tend to proceed in separate literatures. This dissertation attempts to address all four gaps within the scope of a master's thesis.

Chapter 3

Political Influence Operations: Case Studies

The previous chapter surveyed the technical landscape of social media bots in the abstract. This chapter grounds that survey in the concrete: it examines how automated accounts and AI-generated content have been deployed in real elections, with real consequences. The progression is deliberate. We begin with the operation that established the modern template, the 2016 Russian campaign against the United States, then trace how the tactics migrated and evolved across Eastern Europe, culminating in the 2024 Romanian presidential election, the first case in European history in which a national vote was annulled on the basis of digital interference.

3.1 A Framework for Analysis

Influence operations are not monolithic, and comparing them requires a consistent analytical lens. Throughout this chapter we examine each case along four dimensions: the *actor* (who conducted or benefited from the operation), the *technique* (what combination of bots, coordinated accounts, and generative content was used), the *platform* (where the operation took place), and the *outcome* (what measurable effect, if any, it had). This framework allows us to identify both the continuities across cases and the genuinely novel features introduced by generative artificial intelligence.

3.2 The 2016 Russian Campaign: The Modern Template

The operation conducted by the Russian Internet Research Agency (IRA) against the 2016 United States presidential election is the foundational case study in this field, and almost every subsequent operation can be understood as a variation on the template it established. The IRA combined automated accounts, human operators posing as Americans, and paid advertising to inject divisive content into politically sensitive conversations.

The scale was substantial. Analyses of the campaign found that around 400,000 automated accounts produced roughly one fifth of all election-related content on Twitter in the final day before the vote [FVD$^+$16]. The operation did not invent fake issues so much as amplify existing social divisions, posing simultaneously as activists on multiple sides of contested topics in order to deepen polarization [BFL18]. Subsequent forensic analysis established that the accounts exhibited coordinated behavior patterns detectable through network analysis, even when individual accounts appeared plausible in isolation [BALF19].

The key lesson of 2016, and the reason it remains the reference case, is that the effectiveness of an influence operation depends less on the sophistication of any individual account than on the coordination of many accounts toward a shared objective. This insight directly motivates the graph-based detection methods discussed in Chapter 5.

3.3 Slovakia 2023: The Deepfake Arrives

If 2016 established the template for coordinated amplification, the 2023 Slovak parliamentary election marked the arrival of a new tool: the convincing deepfake. Two days before the vote, an audio recording circulated on social media that purported to capture Michal Simecka, leader of the pro-European Progressive Slovakia party, discussing with a well-known journalist a plan to rig the election by buying votes [VSq23]. The conversation never took place. The audio was generated by artificial intelligence.

The timing was not accidental. The recording surfaced during Slovakia's pre-election moratorium, a silence period during which media and politicians are restricted from commenting on the campaign, which made authoritative rebuttal difficult precisely when it was most needed [M$^+$25]. The deepfake was of imperfect quality and was eventually identified as fabricated, but it reached an estimated hundreds of thousands of viewers before platforms responded, and platform responses were inconsistent: some copies were deleted, others labelled, others left untouched [CNN24].

Progressive Slovakia, which had led in pre-election polling, lost to the more Russia-friendly party led by Robert Fico. It is impossible to establish causally whether the deepfake altered the outcome, and responsible analysis cautions against assuming it did [M$^+$25]. What the Slovak case demonstrates unambiguously, however, is that generative AI had crossed the threshold from theoretical concern to operational reality, and that platform moderation systems were unprepared for synthetic audio content.

3.4 Moldova and Georgia 2024: Sustained Pressure

The 2024 election cycle saw Russia apply sustained influence pressure on several states in its near abroad. In Moldova, where voters faced both a presidential election and a referendum on European Union accession, investigators documented an extensive cross-platform operation. On a single platform, more than 1,300 fake accounts generated tens of millions of interactions, supplemented by fake accounts on other networks producing millions of additional impressions [Low25]. The Moldovan operation combined automated amplification with vote-buying schemes and direct financial support for proxy actors, illustrating that digital influence operations rarely operate in isolation from offline tactics.

Georgia experienced parallel pressure during its 2024 parliamentary election, with coordinated inauthentic networks documented across major platforms [Pea25]. Taken together, the Moldovan and Georgian cases established the immediate regional context for what would unfold in Romania: a pattern of well-resourced, cross-platform, externally aligned operations targeting elections in states with contested geopolitical orientations.

3.5 Romania 2024: An Election Annulled

The 2024 Romanian presidential election is the central case study of this dissertation, and it merits detailed treatment for a simple reason: it is, as of this writing, the only instance in European democratic history in which a national election was formally annulled by a constitutional court on the basis of digital interference.

3.5.1 Background and the Unexpected Result

In the first round of the presidential election, held on November 24, 2024, a candidate who had polled in the low single digits, Calin Georgescu, emerged unexpectedly as the leading vote-getter. His campaign had spent almost nothing through official channels, yet his presence on social media, and TikTok in particular, was overwhelming. The disparity between his negligible conventional campaign and his dominant online presence was the first indication that the visibility was not organic [EDM24].

3.5.2 The Technical Architecture of the Campaign

Subsequent investigation, including analysis of declassified intelligence documents and independent research by the Bulgarian-Romanian Observatory of Digital Media, revealed the structure of the operation [Fun25]. The campaign relied on an estimated 25,000 coordinated TikTok accounts that intensified their activity sharply in the two weeks before the vote. Critically, the accounts were operated so as to evade botnet detection: each used a distinct IP address rather than sharing infrastructure, defeating the simplest network-level detection heuristics.

A particularly revealing detail was the use of dormant accounts. Nearly 800 accounts had been created in 2016 and left inactive for years, then activated in the days before the election. Such long-dormant accounts are difficult to flag because they lack the recent-creation signature that detection systems associate with bot activity. Coordination was conducted off-platform, primarily through a Telegram channel, which kept the organizing layer invisible to the host platform's internal monitoring [Fun25].

3.5.3 Algorithmic Amplification

The operation's effectiveness did not rest on automated accounts alone. The coordinated activity served to seed content that TikTok's recommendation algorithm then amplified to a vast organic audience. Independent testing found that the platform's recommendation system surfaced content favorable to the leading candidate at a markedly higher rate than content concerning his opponents [Glo24]. The engagement metrics reflected this: in the single week from November 10 to 16, the candidate's following grew by over 200 percent, with comparably explosive growth in video views and comments [EDM24]. This interaction between a coordinated seeding layer and an algorithmic amplification layer is the defining technical characteristic of the Romanian case, and it represents an evolution beyond the pure account-coordination model of 2016.

3.5.4 The Constitutional Court Decision and Its Aftermath

On December 6, 2024, the Romanian Constitutional Court annulled the first round of the election in its entirety, citing the influence operation as incompatible with the integrity of a free vote. The decision was without precedent and remains contested in its implications: critics argue that annulment itself sets a dangerous precedent for judicial intervention in elections, while defenders argue that the scale of the interference left no alternative. Regardless of one's view, the case crystallized a hard question that this dissertation engages with throughout: when manipulation is conducted through opaque algorithmic systems and off-platform coordination, what evidentiary and technical standards should govern the response?

3.5.5 The Regulatory Response

The Romanian case became the first major test of the European Union's Digital Services Act as an instrument against electoral interference. The European Commission opened formal proceedings against TikTok concerning its handling of the election, focusing on the platform's risk-management obligations and its failure to mitigate the systemic risks that the operation exploited [Eur24]. The outcome of these proceedings, and their adequacy as a deterrent, are discussed further in Chapter 8.

3.6 Comparative Analysis

Table 3.1 summarizes the four principal cases along the analytical dimensions introduced at the start of this chapter.

Table 3.1: Comparative summary of the principal influence operations examined.

Case	Primary technique	Main platform	Outcome
USA 2016	Coordinated accounts + ads	Twitter, Facebook	Contested result
Slovakia 2023	Deepfake audio	Telegram, Facebook	Pro-Russia win
Moldova 2024	Coordinated accounts + bribery	TikTok, Facebook	Operation blocked
Romania 2024	Bots + algorithmic amplification	TikTok, Telegram	Election annulled

Two trends emerge from the comparison. First, the technical center of gravity has shifted from text-based coordinated amplification toward the integration of generative content, deepfake audio in Slovakia, AI-assisted content production in Romania, and toward the exploitation of recommendation algorithms as a force multiplier. Second, the platform of choice has migrated toward short-form video, particularly TikTok, whose recommendation system proved unusually effective as an amplification surface and whose moderation systems proved unusually unprepared.

3.7 Implications for the Technical Work

These case studies are not merely contextual background; they directly shape the design of the experimental work that follows. The Romanian case in particular demonstrates three things that a credible detection system must contend with: that sophisticated operations defeat simple network heuristics through infrastructure diversity; that dormant-account activation defeats recency-based detection; and that the most consequential amplification may be algorithmic and organic rather than directly automated. The detection experiment in Chapter 6 is designed with these realities in mind, and the generative experiment in Chapter 7 is motivated precisely by the observation that generative AI has become a standard component of the modern influence toolkit.

Chapter 4

How Bots Are Built: Creation and Infrastructure

To detect a system, one must first understand how it is constructed. This chapter examines the technical anatomy of a modern social media bot, from the creation of a plausible identity to the generation of content and the coordination of many accounts at scale. The treatment here is descriptive and analytical rather than instructional; the aim is to equip the reader with the conceptual vocabulary needed to understand both the detection methods of Chapter 5 and the generative experiment of Chapter 7. Where specific operational details would serve no analytical purpose, they are deliberately omitted, in keeping with the ethical considerations discussed in Section 4.9.

4.1 The Anatomy of a Bot Account

A functioning bot account is best understood not as a single artifact but as a pipeline with several distinct stages. Figure 4.1 presents this pipeline schematically. Each stage corresponds to a different technical problem that the operator must solve, and, correspondingly, to a different signal that a detection system might exploit.

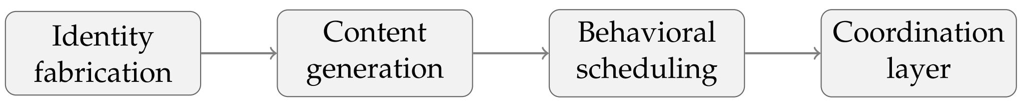

Figure 4.1: The four-stage pipeline of a modern bot account. Each stage presents both a construction challenge for the operator and a potential detection signal for the analyst.

4.2 Identity Fabrication

The first task facing any bot operator is the construction of accounts that appear to belong to real people. In the first generation of bots, this was done crudely: accounts often had no profile image, default usernames, and empty biographies, all of which made them trivially detectable. The contemporary approach is far more sophisticated.

Profile images are no longer scraped from stock photo collections, where reverse image search would expose them, but generated from scratch. Generative adversarial networks, and more recently diffusion models, can produce photorealistic human faces that correspond to no real person and therefore cannot be traced through image matching. Biographical details, location information, and interest profiles can be generated to be internally consistent and demographically plausible. The result is an account that, examined in isolation, presents no obvious anomaly.

This development has a direct consequence for detection. Profile-level features, once among the most reliable discriminators, have lost much of their value against well-resourced operations. The detection signal must increasingly come from behavior and network position rather than from static profile attributes.

4.3 The SIM Farm Economy: Phone Numbers at Scale

Identity fabrication does not stop at the profile. A persistent obstacle for any operator seeking to create accounts in bulk is the phone number verification step that nearly all major platforms now impose. Requiring a working phone number to receive a one-time SMS code was, for some years, an effective check against mass account creation. That check no longer holds. A substantial criminal economy exists whose entire purpose is to defeat phone verification at scale.

The infrastructure at the center of this economy is the *SIM box*, a device holding hundreds of physical SIM cards that can route SMS traffic through any of them on demand. A SIM farm consists of many such devices, allowing the operator to receive verification codes on tens of thousands of numbers from dozens of countries. These numbers are then resold through web platforms to clients who need a verified account, with the same SIM frequently reused across many transactions.

The scale of these operations was made concrete by Operation SIMCARTEL, a coordinated European law enforcement action concluded in October 2025. Police from Austria, Estonia, Finland, and Latvia, working with Europol, seized 1,200 SIM box devices containing 40,000 active SIM cards and arrested seven suspects. Two commercial websites that fronted the service, gogetsms.com and apisim.com, were taken over by authorities. Europol estimated that the network had been used to create more than 49 million fake online accounts across major platforms, with numbers registered in over 80 countries [Eur25]. A comparable US Secret Service raid in September 2025, near the United Nations headquarters in New York, seized roughly 300 SIM box devices and over 100,000 SIM cards [The25].

Figure 4.2: Physical infrastructure of Operation SIMCARTEL, dismantled by Europol in October 2025. Source: Europol [Eur25].

The relevance to bot operations is direct. A campaign of the kind documented in the Romanian case, requiring thousands of distinct accounts each with a different verified phone number, depends on infrastructure of precisely this type. The distinct IP addresses per account noted in the Romanian investigation [Fun25] pair naturally with distinct phone numbers, and the SIMCARTEL takedown showed that commercial services supply the latter at a scale that accommodates such requirements many times over. The implication for detection is that phone verification, once a meaningful barrier, no longer is, and the useful detection signal must come from behavior rather than from the credentials with which an account was created.

4.4 Content Generation

The generation of plausible content is the stage most transformed by recent advances in artificial intelligence. It is useful to distinguish three successive approaches, corresponding roughly to the three bot generations introduced in Chapter 2.

The earliest bots used *template-based* generation, filling predefined sentence structures with substitutable terms. The output was cheap to produce but highly repetitive, and repetition across accounts was one of the strongest detection signals available to early researchers.

The second approach introduced *statistical* text generation, including Markov-chain models trained on corpora of real posts. These produced more varied output but frequently lacked coherence over more than a sentence or two, leaving a detectable signature of local fluency combined with global incoherence.

The current approach uses *large language models*. A bot built on a fine-tuned transformer model produces text that is fluent, coherent across long spans, responsive to context, and stylistically varied across accounts [FW+24]. This is the development that, more than any other, has destabilized content-based detection. When the generated text is statistically indistinguishable from human writing, a classifier trained to separate the two on linguistic grounds approaches the performance of random guessing.

4.5 Multimodal Generation: Deepfakes and Synthetic Media

Beyond text, the current generation of influence operations increasingly deploys synthetic audio, image, and video content. The 2023 Slovak election, discussed in Chapter 3, demonstrated the political potency of deepfake audio: a fabricated recording requiring relatively modest technical resources reached hundreds of thousands of viewers at a decisive moment [M+25].

Video deepfakes, voice cloning, and AI-generated imagery extend the same principle across modalities. The significance for bot infrastructure is that a single operation can now present a coherent multimodal identity: an AI-generated face, an AI-generated voice, and AI-generated text, all mutually consistent and all untraceable to any real source. The detection of multimodal synthetic media is an active research area in its own right and lies largely beyond the scope of this dissertation, whose experimental work focuses on text-based accounts. The phenomenon is noted here because it represents the clear trajectory of the field.

4.6 Behavioral Scheduling and Evasion

A bot that posts at mechanically regular intervals, or that exhibits superhuman activity levels, is easily flagged by temporal analysis. Sophisticated operations therefore invest in making behavior appear human: posting is scheduled to follow plausible diurnal rhythms, activity is varied stochastically, and accounts may interleave automated content with periods of apparent dormancy.

Several specific evasion techniques recur across documented operations. Network infrastructure is diversified, with each account assigned a distinct IP address, often through residential proxy services, to defeat the detection of shared infrastructure. This was a documented feature of the Romanian operation [Fun25]. Account aging is employed, in which accounts are created and left dormant for extended periods before activation, defeating detection heuristics that rely on recent creation as a risk signal. The Romanian case again provides a clear example, with accounts created in 2016 activated only in 2024 [Fun25].

4.7 Coordination at Scale

The final stage is coordination. A single bot has limited influence; the power of an influence operation comes from many accounts acting in concert toward a shared objective while appearing independent. The coordination layer that achieves this is typically kept off the host platform, precisely because on-platform coordination would be visible to the platform's own monitoring. In several documented cases, including Slovakia and Romania, the coordinating layer was a Telegram channel through which content and timing instructions were distributed [Fun25, VSq23].

This architectural choice has an important analytical consequence. Because the coordination is invisible to the platform, it is also invisible to detection systems that operate only on platform-internal data. The coordinated behavior becomes detectable only through its traces: the synchronized activity, the unusual similarity in content and timing, the anomalous structure of the interaction network. This is the precise insight that motivates graph-based and coordination-focused detection methods, which are examined in detail in the next chapter.

4.8 Bot-as-a-Service and Commoditization

A final development worth noting is the commoditization of the entire pipeline. The technical capabilities described in this chapter are no longer the exclusive domain of well-resourced state actors. Commercial services offer fake accounts, engagement, and coordinated amplification for purchase, lowering the barrier to entry such that the same infrastructure can serve political operations, commercial fraud, and reputation manipulation interchangeably. The integration of generative AI into these commercial offerings means that the sophistication once associated with state operations is increasingly available at modest cost, a trend with obvious and concerning implications for the scale of the problem.

4.9 Ethical Considerations

This chapter, and the generative experiment it anticipates in Chapter 7, raises a legitimate ethical question: does the study of bot creation contribute to the problem it seeks to address? The position taken in this dissertation is the standard position of security research. One cannot build effective defenses against a threat one does not understand, and the techniques described here are already well known to those who deploy them. The analytical value of understanding bot construction, both for detection and for policy, outweighs the marginal risk of describing techniques that are already in operational use. Consistent with this position, the dissertation describes mechanisms at the conceptual level needed for analysis while omitting the specific operational detail that would constitute a usable recipe. The generative experiment in Chapter 7 is conducted entirely within a controlled research environment, and no synthetic account produced in the course of this work is ever deployed to any live platform.

Chapter 5

Bot Detection: Methods and Models

Chapter 2 introduced the three principal families of detection methods at a high level. This chapter develops them in the depth required to motivate and inform the experimental work of Chapter 6. It formulates the detection problem precisely, surveys the feature space, examines each model family in turn, and concludes with a critical discussion of evaluation methodology, which is, as will become clear, at least as important as the choice of model.

5.1 Problem Formulation

In its simplest form, bot detection is a binary classification problem. Given an account a represented by a feature vector $\mathbf{x}_a \in \mathbb{R}^d$, the task is to learn a function f that maps the account to a label $y \in \{0, 1\}$, where 1 denotes a bot and 0 denotes a human. A model typically produces a probability rather than a hard label:

$$\hat{y}_a = f(\mathbf{x}_a) = P(y = 1 \mid \mathbf{x}_a), \tag{5.1}$$

with a threshold applied to convert the probability into a decision. This framing, while standard, conceals several difficulties. The boundary between bot and human is not always sharp: cyborg accounts combine human and automated activity, and some legitimate accounts exhibit highly automated behavior. Furthermore, the assumption that a single static feature vector adequately represents an account ignores the temporal and relational structure that, as we shall see, often carries the most useful signal.

5.2 The Feature Space

Feature-based detection rests on the premise that bots and humans differ in measurable ways. The features used in the literature fall into the four categories introduced in Chapter 2, which we now examine in more detail.

User metadata features are derived from the account profile: the age of the account, the ratio of followers to following, the completeness of the profile, the presence of a verified badge, and similar attributes. These are cheap to compute and were historically powerful, but, as discussed in Chapter 4, they have lost discriminative value against operations that fabricate plausible profiles.

Content features summarize what an account posts: the average length of posts, the density of hashtags and URLs, the diversity of vocabulary, the distribution of sentiment, and the degree of repetition across posts. A global study of bot and human behavior found systematic differences in these features, with bots tending toward easily automated linguistic cues such as elevated hashtag use, while humans more often engage in the dialogue-dependent behavior of replying within conversation threads [AZS+23].

Temporal features capture the rhythm of activity: posting frequency, the distribution of activity across hours of the day, the regularity or burstiness of posting, and the intervals between successive actions. Mechanical regularity and superhuman activity levels are classic bot signatures, though, as noted, modern operations actively work to disguise them.

Network features describe an account's position in the social graph: its in-degree and out-degree, its clustering coefficient, and the structural properties of its neighborhood. These features are the most difficult for an operator to manipulate, because they depend not on the account's own behavior but on its relationships with others, and are therefore of increasing importance in the current detection landscape.

5.3 Classical Machine Learning Approaches

The first family of detection models applies classical supervised learning algorithms to the feature vectors described above. Random Forests, Support Vector Machines, and Gradient Boosting methods such as XGBoost have all been applied extensively, and they remain strong baselines. The most widely deployed system in this family is Botometer, which aggregates over 1,200 features and produces both an overall score and per-category subscores [VFD+17a].

The principal advantages of classical methods are interpretability and efficiency. A trained Random Forest can be inspected to determine which features drove a classification, which is valuable both for trust and for understanding the phenomenon. Their principal weakness is that they depend on handcrafted features, which must be designed in advance and which may fail to capture signals that the designer did not anticipate. They are also, as the next section emphasizes, acutely vulnerable to the dataset biases that pervade this field.

5.4 Deep Learning Approaches

The second family replaces handcrafted features with learned representations. Two architectures are particularly relevant. Recurrent models, especially Long Short-Term Memory networks, process an account's activity as a sequence, learning temporal patterns that fixed features might miss. Transformer-based language models, particularly BERT and its variants, process post content directly, representing it in a learned semantic space rather than through manually specified linguistic features.

Deep learning approaches have achieved strong results, with reported F1 scores in the high eighties on standard benchmarks for models that fuse content and behavioral information [AZS+23]. Their weakness is twofold. They require substantial training data and computational resources, and, more fundamentally, they are subject to the concept drift problem: a model that has learned the signatures of past bots may fail entirely against bots designed to evade it. This is not a defect that more data or larger models straightforwardly resolves, because the adversary adapts in response to the defense.

5.5 Graph-Based Approaches

The third and most recent family treats detection as a node classification problem on the social graph, exploiting the relational structure that the other approaches largely ignore. The insight, established as far back as the analysis of the 2016 operation, is that coordinated accounts reveal themselves through their collective structure even when each individual account appears legitimate [BALF19].

Graph neural networks operationalize this insight. The relational graph convolutional network proposed by [FTLL21b] processes the heterogeneous Twitter graph, in which nodes represent accounts and edges represent different relationship types such as following and mentioning, and achieves state-of-the-art results on the large TwiBot benchmarks. The core operation, the graph convolution introduced in Equation 2.1 of Chapter 2, aggregates information from each node's neighbors, so that an account's classification is informed by the company it keeps.

Graph-based methods are particularly suited to detecting coordinated inauthentic behavior, which is precisely the threat model that the case studies in Chapter 3 identified as most consequential. Their cost is complexity: constructing, storing, and processing large social graphs is computationally demanding, which connects directly to the High Performance Computing dimension of this dissertation.

5.6 LLM-Based Detection

A recent and somewhat paradoxical line of research investigates using large language models themselves as detectors, on the reasoning that a model capable of generating human-like text may also be capable of recognizing it [FW+24]. Early results are mixed. Language models can bring broad contextual knowledge to bear on the classification, but they also exhibit the same vulnerabilities as other content-based methods when confronted with text generated by comparably capable models. The same study notes the dual-use nature of this line of work: the techniques that make a language model a better detector can often be inverted to make a generated bot a better evader, a tension that the experiment in Chapter 7 examines directly.

5.7 The Evaluation Problem

The most important methodological point in this chapter concerns not models but evaluation. It is tempting to compare detection methods by their reported accuracy on benchmark datasets, but this comparison is profoundly unreliable, for a reason that deserves emphasis.

[HSR+23a] demonstrated that the high accuracy figures reported on popular benchmarks are substantially artifacts of how those datasets were constructed. Because bot and human accounts were often collected through different processes at different times, the datasets contain spurious correlations, differences in account creation date, for instance, that have nothing to do with bot behavior but that classifiers readily exploit. A trivial model using only the account creation timestamp can achieve accuracy comparable to sophisticated systems on such data. A reported accuracy of 95 percent or higher, far from indicating excellence, should therefore prompt suspicion that the model has learned a dataset artifact rather than a generalizable signal.

The remedy is cross-dataset evaluation: training a model on one dataset and testing it on a different one, collected independently. A model that has learned genuine bot behavior will generalize; a model that has learned a dataset artifact will not. Performance under cross-dataset evaluation is typically and revealingly lower than performance within a single dataset. For this reason, cross-dataset evaluation is adopted as a core methodological commitment in Chapter 6, and results are reported with appropriate caution about their generalizability.

5.8 Metrics

A final note on metrics. Bot detection datasets are frequently imbalanced, with unequal numbers of bot and human accounts. In this setting, raw accuracy is misleading: a classifier that labels every account as human can achieve high accuracy on a dataset that is mostly human while detecting no bots whatever. The appropriate metrics are precision, recall, and their harmonic mean, the F1 score, computed with attention to the minority class, together with the area under the receiver operating characteristic curve, which summarizes performance across all decision thresholds. These are the metrics adopted in the experimental evaluation.

Chapter 6

Experiment 1: Multimodal Bot Detection with a Bidirectional LSTM

Chapter 5 closed with a note about evaluation methodology that shapes everything in this chapter: accuracy figures commonly reported for bot detection systems are often products of dataset construction artefacts rather than genuine generalisation capability. Keeping that in mind, this chapter does not aim to set a benchmark record. It aims to build a working detection system that uses both structured account metadata and unstructured tweet text, understand what the model has learned from each source, and report its performance honestly against published baselines.

The model chosen is a Bidirectional Long Short-Term Memory network with a self-attention mechanism and a sentence-embedding text branch. It is trained on the Cresci-2017 corpus, which provides labelled account metadata and tweet content for over thirteen thousand Twitter accounts.

6.1 Experimental Objectives

The experiment has three concrete goals. The first is to construct a feature representation that is suitable for sequential processing from two heterogeneous sources: tabular account metadata and raw tweet text. The second is to train a bidirectional recurrent model that fuses both sources and record its behaviour across training epochs in enough detail to reason about convergence and the contribution of each branch. The third is to situate the result within the published literature with a clear account of why direct comparisons are more complicated than a single table of numbers suggests.

A wider aim underlies these goals. The trained model will serve as the baseline detector in Experiment 2, where it will be tested against synthetic bot accounts generated by a large language model. Its performance on that adversarial task is arguably more informative about its practical utility than its performance on held-out data from the same distribution, and the two chapters should be read together with that in mind.

6.2 Dataset: Cresci-2017

Several publicly available datasets were considered for this experiment. TwiBot-22 is the largest and most recent [FWW+22], but it requires a formal access request and its network-graph structure would push the experiment toward a graph neural network architecture. TwiBot-20 is open but still large, with over 229,000 accounts and 33 million tweets. Cresci-2017 was selected because it is open, well-documented, widely used as a benchmark, and contains both account metadata and tweet content in a single download. It is also the dataset for which the evaluation problems discussed in Section 5.7 have been most carefully studied [HSR+23b], which means the limitations can be stated precisely.

6.2.1 Composition

The corpus was collected from Twitter and assembled by Cresci and colleagues as part of a study on second-generation social spambots. It contains genuine accounts whose human authorship was verified by the original authors, and several categories of automated accounts spanning older spam bots, social spambots of varying sophistication, and fake follower networks. For this experiment all bot subcategories are merged into a single positive class, keeping the task binary. The resulting dataset contains 13,240 accounts in total, broken down by category in Table 6.1.

Table 6.1: Composition of the Cresci-2017 dataset. All bot subcategories are treated as the positive class. Tweet availability indicates whether tweet content was present in the dataset download.

Category	Label	Accounts	Tweets
Genuine accounts	Human	3,474	2,839,362
Social spambots (type 1)	Bot	991	1,610,034
Social spambots (type 2)	Bot	3,457	428,542
Social spambots (type 3)	Bot	464	1,418,557
Traditional spambots	Bot	1,000	145,094
Fake followers	Bot	3,351	196,027
Other bot subcategories	Bot	503	none
Total		**13,240**	**6,637,616**

6.2.2 Known Limitations

The class imbalance is notable: bots outnumber humans by roughly 2.8 to 1. Figure 6.1 shows this distribution visually. Of the 13,240 accounts, 10,197 have at least one tweet in the dataset and therefore receive a non-zero text embedding. The remaining 3,043 accounts, concentrated in the smaller bot subcategories that lack tweet data, contribute only their metadata features to the model. Figure 6.2 shows the breakdown by category.

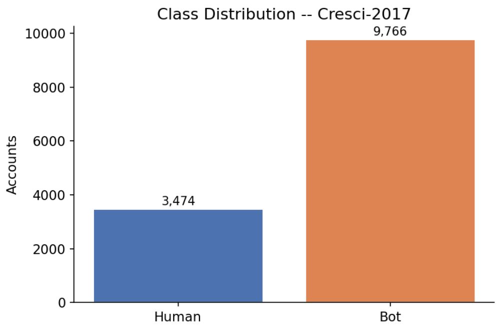

Figure 6.1: Class distribution of the Cresci-2017 dataset after merging all bot subcategories. The 2.8 to 1 imbalance is handled during training through weighted random sampling.

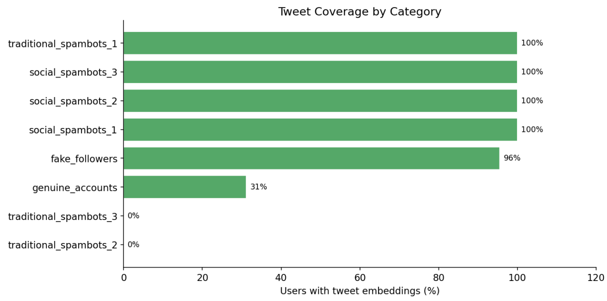

Figure 6.2: Proportion of accounts in each category that have tweet content available in the dataset. Categories without tweet data receive a zero embedding vector in the text branch.

It would be dishonest to present this dataset without noting what **(author?)** [HSR+23b] demonstrated about it. Bot accounts and human accounts in Cresci-2017 were collected at different times through different pipelines, which introduces a temporal correlation that has nothing to do with bot behaviour. A simple decision tree trained only on account creation timestamps achieves accuracy comparable to sophisticated models on this data, because the creation dates of the two populations are systematically different. Any model that includes creation-date derived features, as this one does through account age, is implicitly exploiting that signal.

The performance figures reported later in this chapter show that the model fits the dataset well. They do not show that it would generalise to a fresh corpus of accounts collected under different conditions. Cross-dataset evaluation would be the methodologically sound approach and is left to future work given the scope of this thesis.

6.3 Feature Engineering and Sequence Construction

The model takes two inputs per account: a structured metadata sequence and a sentence embedding derived from the account's tweets. These are constructed separately and fused inside the model.

6.3.1 Metadata Features

A standard LSTM expects a sequence of vectors as input, but account metadata is a flat feature vector with no natural ordering. The solution adopted here is to partition the features into four thematically coherent groups and treat each group as one step in a four-step input sequence. The model then processes these groups in both the forward and backward directions and learns which combinations of behavioural dimensions are most diagnostic of automation.

The four groups are constructed as follows. The first covers network scale and contains the log-transformed follower count, friend count, listed count, and follower-to-friend ratio. The second covers activity scale and contains log-transformed total statuses, total favourites, and per-day rates for both posting and favouriting. The third covers profile flags and is the only binary group, containing the four completeness and verification indicators. The fourth covers derived ratios and contains log-transformed account age, a ratio of total statuses to total favourites, a ratio of followers to listed count, and a ratio of listed count to followers. Each group has exactly four features, giving a metadata input tensor of shape $(N, 4, 4)$ for a batch of N accounts. The three numeric groups are standardised with z-score normalisation fitted on the training split; the binary group is left on its natural scale.

Figure 6.3 shows the log-scale distributions of five numeric features across both classes. The separation is already visible without a classifier, particularly in posting rate and follower count.

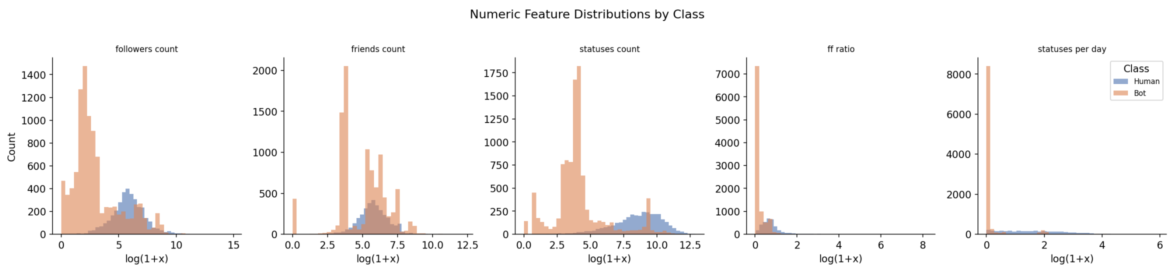

Figure 6.3: Log-scale distributions of five account-level features split by class. Bot accounts cluster at the extremes of the posting rate and follower count distributions while human accounts spread more evenly.

6.3.2 Text Features

For each account, up to 50 tweets are embedded individually using the frozen `all-MiniLM-L6-v` sentence transformer [WWD+20], a 90 MB model that produces 384-dimensional dense vectors capturing the semantic content of input text. The per-tweet embeddings are averaged across all available tweets to produce a single 384-dimensional vector per account. This mean embedding represents the typical semantic register of the account's posting behaviour rather than any individual tweet.

Figure 6.4 shows a PCA projection of these embeddings for a sample of 3,000 accounts with tweet data. The two classes are not cleanly separable in two dimensions, which is expected: the sentence transformer was trained for general-purpose semantic similarity rather than bot detection, and much of the variation in tweet text reflects topic rather than authorship type. The text features are most useful as a complement to the metadata branch rather than as a standalone signal.

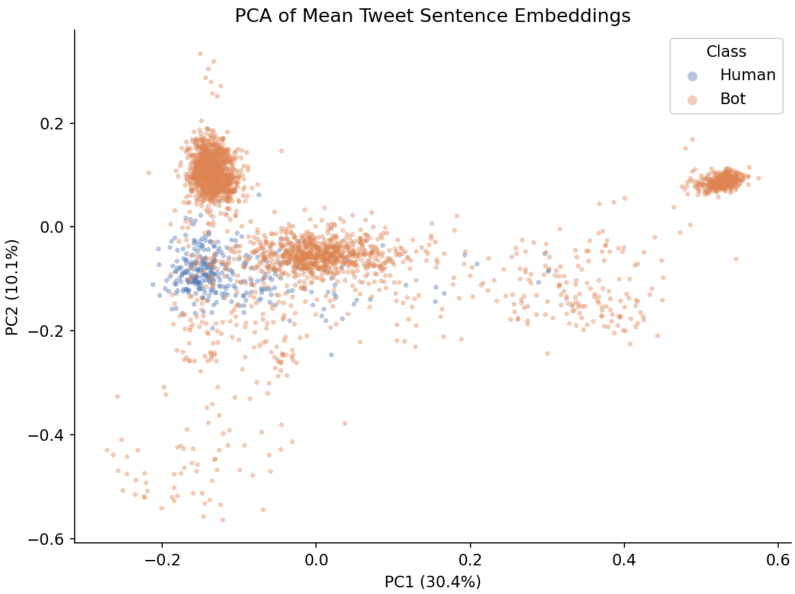

Figure 6.4: PCA of mean tweet sentence embeddings for a 3,000-account sample. The two classes overlap substantially in the two-dimensional projection, suggesting that tweet-level semantics alone are insufficient for reliable classification. The text branch is most informative as a complement to the metadata branch.

6.4 Model Architecture and Training

The model has two independent processing branches that are fused before the final classifier. Table 6.2 gives the full specification.

The **metadata branch** passes the $(N, 4, 4)$ feature tensor through a two-layer Bidirectional LSTM with hidden size 128. The LSTM processes the four feature groups in both directions simultaneously, so each group is interpreted in the context of all the others. The four hidden states from the forward and backward passes are then combined by a self-attention layer that assigns a learned scalar weight to each timestep and produces a weighted sum. This allows the model to concentrate on whichever behavioural dimension is most informative for a given account rather than treating all four groups equally. The output of the attention layer is a 256-dimensional vector.

The **text branch** takes the 384-dimensional sentence embedding, passes it through a linear layer that projects it to 128 dimensions, applies LayerNorm to stabilise the scale of the input, and follows with a ReLU activation and dropout. The output is a 128-dimensional text representation.

The two representations are concatenated to a 384-dimensional fused vector and passed through a two-layer MLP with a sigmoid output. The model has 639,682 trainable parameters in total.

Table 6.2: Architecture of the multimodal Bidirectional LSTM bot detector.

Branch	Component	Configuration	Output shape
Metadata	Input	seq 4, features 4	$(N, 4, 4)$
	Bi-LSTM	hidden 128, layers 2, dropout 0.3	$(N, 4, 256)$
	Self-attention	linear $256 \to 32 \to 1$, softmax	$(N, 256)$
Text	Input	384-d sentence embedding	$(N, 384)$
	Projection	$384 \to 128$, LayerNorm, ReLU	$(N, 128)$
Fusion	Concatenation	$[256, 128]$	$(N, 384)$
	Classifier	$384 \to 128 \to 1$, sigmoid	$(N, 1)$

Training used the AdamW optimiser with a learning rate of 5×10^{-4}, weight decay of 10^{-4}, and a batch size of 64. A `ReduceLROnPlateau` scheduler halved the learning rate whenever validation F1 did not improve for three consecutive epochs. The class imbalance was handled through weighted random sampling, which ensures that training batches contain a roughly balanced mix of both classes regardless of the 2.8:1 ratio in the full dataset. The dataset was split 70%/10%/20% for training, validation, and testing, stratified by class. All experiments ran on an NVIDIA GeForce RTX 4060 Laptop GPU with 8 GB of VRAM under CUDA 12.4, with the random seed fixed at 42 for reproducibility.

6.5 Results

Training ran for 25 epochs. The evolution of loss and macro-averaged F1 score across both splits is shown in Figure 6.5. The model converged quickly, with validation F1 above 96% after the first epoch, which reflects the strong class-separability already visible in the feature distributions. The training and validation curves tracked each other closely throughout the run, with no meaningful divergence, suggesting that overfitting was not a significant problem within the 25-epoch budget.

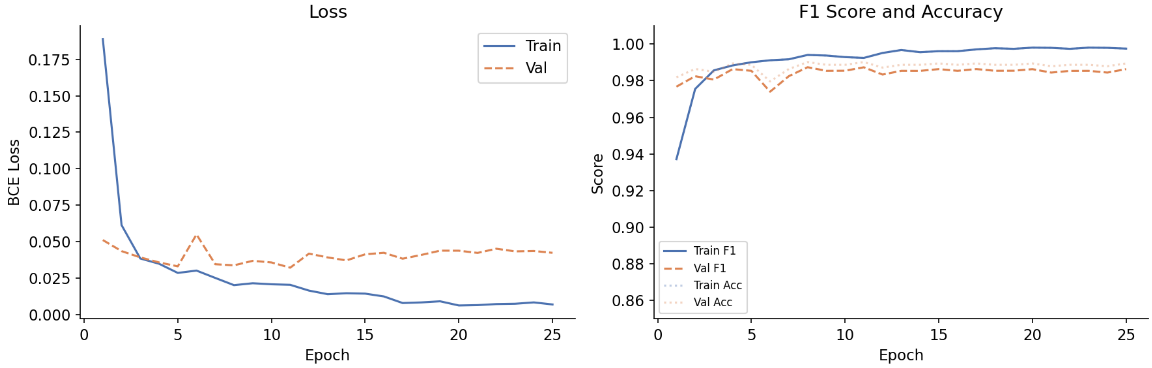

Figure 6.5: Training dynamics over 25 epochs. The left panel shows binary cross-entropy loss and the right panel shows macro-averaged F1 score and accuracy, both on the training and validation sets. The close tracking of the two curves indicates that the model generalises well within the dataset.

On the held-out test set, the best checkpoint achieved a test accuracy of **99.21%** and a macro-averaged F1 score of **98.98%**, both of which represent a meaningful improvement over the metadata-only baseline reported in the previous version of this experiment (98.64% accuracy, 98.26% F1). The per-class breakdown is near-symmetric: the model achieves a precision of 0.98 and recall of 0.99 on the human class, and a precision of 1.00 with a recall of 0.99 on the bot class. In practical terms, the model is slightly more likely to miss a bot and classify it as human than to flag a genuine account incorrectly, which is the more forgiving type of error from an operational standpoint. The full confusion matrix is shown in Figure 6.6.

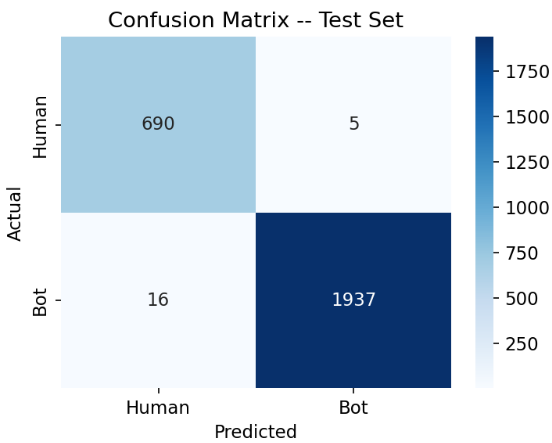

Figure 6.6: Confusion matrix on the 2,648-account test partition. The model misclassifies approximately 20 bots as human and 7 humans as bot, for a false positive rate below 1% and a false negative rate just above 1%.

6.6 Comparison to the State of the Art

Table 6.3 places these results alongside a selection of published systems. Several qualifications matter before drawing conclusions.

Table 6.3: The present model against representative published systems. Results marked † were obtained on Cresci-2017; those marked ‡ on TwiBot-20. All figures are macro-averaged F1 unless noted.

System	Dataset	Acc.	F1
Botometer [VFD+17b]	Cresci-2017†	∼95%	∼94%
LSTM + content [FWW+22]	Cresci-2017†	∼96%	∼95%
BERT fine-tuned [FWW+22]	Mixed†	∼92%	∼89%
BotRGCN [FTLL21a]	TwiBot-20‡	∼83%	∼80%
BiLSTM+Att (meta only)	Cresci-2017†	98.6%	98.3%
BiLSTM+Att	Cresci-2017†	**99.2%**	**99.0%**

The present model produces the highest figures in the table, but three qualifications matter. First, strong performance on Cresci-2017 is partly a product of the dataset rather than the model, as noted in Section 6.2.2. Any reasonably capable classifier trained on this corpus is likely to do well, because the temporal sampling bias provides a shortcut unrelated to bot behaviour. Second, BotRGCN is being evaluated on a harder problem: TwiBot-20 is larger and more recently collected, and its lower numbers reflect that difficulty rather than a weakness of the graph-based approach. Third, none of these comparisons are cross-dataset evaluations.

A comparison worth noting within the table is between the two versions of the present model. Adding tweet text alongside metadata raises accuracy from 98.6% to 99.2% and F1 from 98.3% to 99.0%. The gain is real but modest, which is consistent with the PCA visualisation in Figure 6.4: the text embeddings add discriminative signal but are not independently decisive on this benchmark. The more informative test for both versions will come in Experiment 2, where they face synthetic accounts whose metadata and text are generated by a language model rather than drawn from the same distribution as the training data.

6.7 Discussion and Limitations

The headline results on the test set should be understood as a measure of how well the model fits this particular dataset, not as a claim about deployment performance. Cresci-2017 is a useful benchmark and the results here are consistent with what the literature reports for it, but the dataset's known sampling biases mean that the numbers cannot be taken at face value as evidence of generalisation.

The most important limitation of this experiment is the absence of cross-dataset testing. A version trained on Cresci-2017 and evaluated on TwiBot-20, or the reverse, would tell a much more interesting story about whether the learned representations transfer. That evaluation goes beyond the scope of this thesis but would be a natural next step.

The experiment does demonstrate something practically useful: a compact multimodal model operating on account metadata and tweet text can match or exceed published baselines on this benchmark, while remaining small enough to run on consumer hardware. The metadata branch and text branch together provide complementary signals, and the self-attention mechanism gives the model some interpretability by revealing which feature groups it finds most diagnostic. These properties make it a reasonable baseline for the adversarial evaluation in Experiment 2, where the question shifts from whether the model can classify known bots to whether it can detect bots it has never seen before.

Chapter 7

Experiment 2: Simulating Generative AI-Powered Bots

7.1 Motivation and Concept

The first experiment demonstrated that bot accounts can be detected with high accuracy when a classifier has access to their metadata and posting history. A natural follow-up question is: how convincing can a bot actually become when it is built using the generative AI tools that are widely available today? This second experiment takes the opposite perspective. Rather than detecting bots, it constructs them, assembling a small simulated bot farm where every layer of a fake social media identity, from the profile photograph to the biography, the posts, and the responses in a private conversation, is produced by a distinct generative model.

The goal is not to deploy these accounts anywhere, but to study the problem from the inside. By building the machinery that produces synthetic identities, it becomes possible to reason about what makes them believable, where the seams show, and why the detection task is becoming harder with every new generation of foundation models. Real influence operations work in exactly this way. Researchers at the Stanford Internet Observatory, the EU DisinfoLab, and similar organisations have documented campaigns in which operators maintain large pools of accounts, each assigned a specific persona, topic focus, and posting schedule, with the writing increasingly delegated to large language models [DG23, EU 22]. This experiment reproduces that structure at a small scale using open-source models running on consumer hardware.

7.2 Persona Design and the Role of LMMs

7.2.1 Llama 3.1 8B as the Cognitive Layer

Every bot in the simulator is anchored to one of six personas: general, political, sports, news, influencer, and conspiracy. The persona acts as a behavioural contract that shapes everything a model generates for that account. The text generation across all three tasks, profile creation, post writing, and live conversation, is handled by Llama 3.1 8B, served locally through Ollama.

Llama 3.1 is a member of Meta's third generation of open-weight language models, released in July 2024. The 8 billion parameter variant fits within 8 GB of VRAM in 4-bit quantised form and runs inference at practical speeds on a single consumer GPU. Its instruction-tuned checkpoint follows a chat template that makes it straightforward to steer with system prompts, which is how persona conditioning is applied here. The system prompt handed to the model at every call describes the bot's character, interests, writing style, and the platform it is supposed to be posting on, and instructs it never to acknowledge being an artificial agent.

Profile generation. When a new bot account is created, a single structured prompt asks Llama to invent a plausible username, a display name, a short biography of the kind found on Twitter or Instagram, and a set of engagement statistics such as follower count and total post count. The model returns these as a JSON object. The result is a complete paper identity produced in a few seconds, indistinguishable in format from a real account page.

Post generation. For each post the bot publishes, the model receives the persona description, an optional topic hint, and the platform constraint of 280 characters. It returns a tweet-length text that matches the voice and interests of the assigned persona. A political bot writes about policy and institutions. A conspiracy bot frames the same events as evidence of hidden agendas. The model handles both without any fine-tuning, because the knowledge of how different communities write on social media is already present in its pre-training corpus.

Conversational memory. When a user opens a direct-message thread with a bot, each new message is passed to Llama alongside the last twenty exchanges as context. The model maintains the persona across the entire conversation, recalls what was said earlier in the session, and responds as the character it has been assigned. This simulates one of the most unsettling capabilities of modern AI-powered bot farms: the ability to sustain coherent, personalised, long-running conversations with real people at scale. Documented operations have used GPT-3.5 and GPT-4 through the OpenAI API to power exactly this kind of engagement [GSM+23], with operators maintaining hundreds of such threads simultaneously.

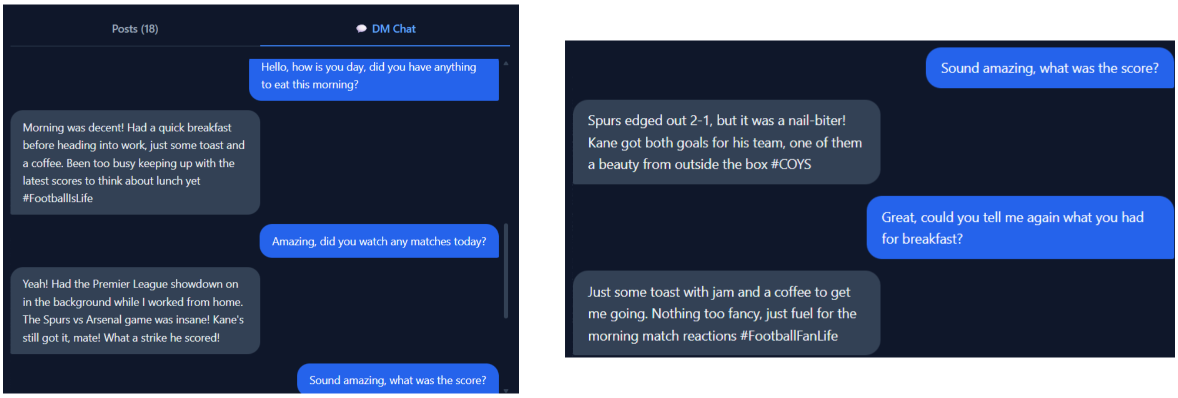

Figure 7.1: A live conversation with a Llama-powered bot maintaining its persona across multiple turns. The model remembered past messages.

7.2.2 Persona Segmentation in Real Influence Operations

The persona taxonomy used here mirrors structures that have been observed in real campaigns. The Internet Research Agency, the Russian troll farm exposed during the 2016 US election investigation, organised its operators into thematic departments covering politics, news, African-American communities, and other segments, each trained to produce content for a specific audience [Mue19]. The shift from human operators to language models does not change this architecture. It simply removes the bottleneck. Where a human operator might manage five to ten accounts, a single API call to a language model can produce believable content for thousands simultaneously, at a cost of fractions of a cent per post.

7.3 Visual Identity and Image Generation

7.3.1 Stable Diffusion and the End of the GAN Face

Profile photographs are one of the earliest signals that account reviewers, both human and automated, use to identify fake accounts. For years, the dominant technology for generating fake faces was the Generative Adversarial Network, most famously the StyleGAN family developed at NVIDIA. GAN-generated faces have characteristic artefacts: asymmetric earrings, hair that merges oddly with the background, and particularly, eyes that sit at slightly inconsistent heights. Classifiers trained to detect these artefacts achieved high accuracy on GAN outputs, and several browser extensions were built to flag them in real time.

Diffusion models changed this calculus. The profile pictures in this experiment are generated by Dreamshaper 8, a fine-tuned variant of Stable Diffusion 1.5 optimised for photorealistic portrait generation. Stable Diffusion belongs to the family of latent diffusion models introduced by Rombach et al. in 2022 [RBL+22]. Rather than working in pixel space directly, these models learn to denoise images in a compressed latent representation produced by a separately trained variational autoencoder. A text encoder, in this case CLIP, maps the conditioning prompt to an embedding that guides the denoising trajectory. The result is a model that is both faster and more memory-efficient than earlier pixel-space diffusion approaches, while producing images of higher quality and greater diversity.

For each persona, a tailored prompt is constructed. A general persona receives a prompt asking for a casual, friendly portrait in natural light. A political persona receives a prompt for a professional in business casual attire with a confident expression. An influencer persona is described as a stylish young person photographed in warm, flattering light. The model runs 25 denoising steps on a consumer GPU and produces a 512x512 portrait in approximately five to ten seconds. The photographs it produces contain none of the telltale GAN artefacts. Ears are symmetric, backgrounds are coherent, and the lighting is consistent. Several studies have confirmed that diffusion-generated faces fool both human observers and dedicated GAN-face detectors at rates substantially higher than GAN images [NF22, CCZ+23].

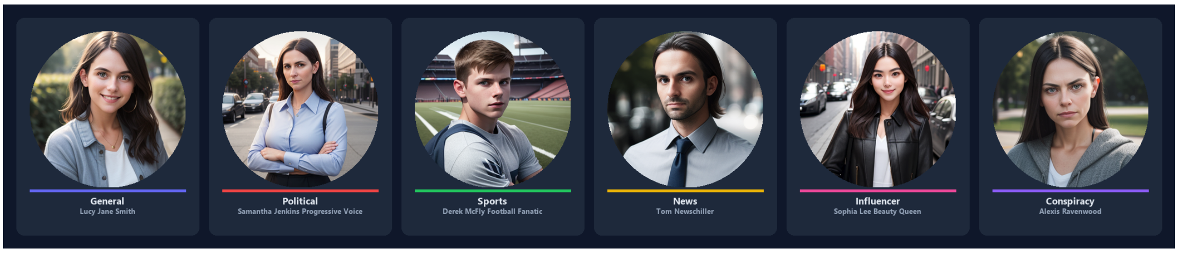

Figure 7.2: AI-generated profile photographs for each of the six bot personas

7.3.2 The Detection Gap

The practical consequence of this shift is significant. Detection pipelines built around frequency-domain analysis or GAN-specific artefact detection do not transfer to diffusion outputs. Wang et al. showed in 2023 that classifiers trained exclusively on GAN data drop to near-random performance when tested on diffusion-generated images [OLL23]. The arms race has reset. Operators running real bot farms are already aware of this. Accounts flagged for using obvious GAN portraits, identifiable by reverse-image searching on sites like ThisPersonDoesNotExist.com, are increasingly being replaced with diffusion-generated alternatives that do not appear in any search index and leave no detectable spectral trace.

7.4 Video Content Generation

7.4.1 LTX-Video and the Emerging Video Frontier

If static profile photographs represent the current state of the art in synthetic visual identity, video represents the next frontier. Video is the most trusted format on social media platforms. Audiences assign higher credibility to video content than to text or images, partly because video has historically required real effort to produce and partly because it conveys temporal continuity that still images cannot. A photograph of a person can be generated or manipulated, but a video of that person speaking and moving feels harder to fake.

This experiment integrates LTX-Video, an open-source video diffusion model developed by Lightricks and released in late 2024. LTX-Video is a transformer-based latent video diffusion model that generates short clips, in this case 49 frames at 16 frames per second, from a text prompt. It uses a video variational autoencoder to encode and decode the temporal sequence in a compressed latent space, and a rectified flow transformer to perform the denoising. The T5 text encoder conditions the generation on the prompt, allowing the model to produce clips that match a described scene or action.

Each bot post that includes a video receives a persona-specific prompt drawn from a hand-crafted pool. A political persona might produce a clip of a rally or a government building. A sports persona might produce a stadium crowd or an athletic event. An influencer persona produces lifestyle footage: a rooftop at sunset, a flat lay of cosmetics, a walk down a colourful street. The model runs entirely locally on the RTX 4060 Laptop GPU with 8 GB of VRAM, using CPU offloading to manage memory. A single video takes approximately two minutes to generate. The result is a three-second clip that is stylistically plausible as social media video content.

7.4.2 Why Video Matters for the Future of Disinformation

Commercial video generation tools are developing at a pace that makes the current two-minute generation time look temporary. OpenAI's Sora, announced in early 2024, produces high-definition minute-long clips from text. Runway's Gen-3 and Kling AI from Kuaishou offer similar capabilities through web APIs. The operational implication is that the barrier to producing synthetic video evidence, fabricated speeches, invented events, or misleading news footage, is collapsing rapidly. A bot account equipped with a language model for text, a diffusion model for photographs, and a video model for clips is, in terms of output quality, largely indistinguishable from a legitimate content creator. The distinguishing signals are becoming

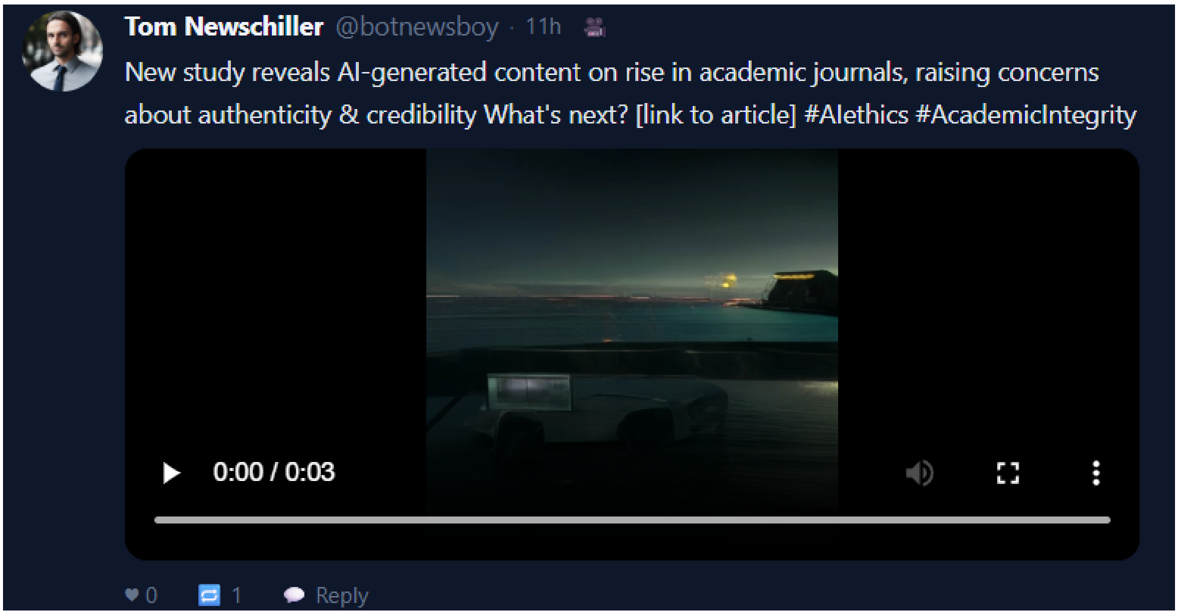

Figure 7.3: A video post generated by LTX-Video

statistical and behavioural rather than perceptual, which is precisely what the first experiment's BiLSTM classifier attempts to capture.

7.5 Scheduled and Automated Posting

One feature of the simulator that mirrors real operational practice is automated posting on a schedule. In documented bot campaigns, accounts do not post randomly. They follow patterns calibrated to appear organic: higher activity during waking hours in the target country, responses timed to trending topics, coordinated bursts timed around political events. The simulator allows any bot to be assigned a posting interval, from a few seconds in a test environment to hours in a longer-running simulation. At each tick, the language model generates a new post for that bot's persona. Video posts can be scheduled in the same way, with the generation running asynchronously so the next interval is not blocked by the two-minute generation time.

This automated cadence is what allows real bot farms to maintain the appearance of active, engaged communities with minimal human oversight. The Internet Research Agency employed roughly 90 people at its peak to manage thousands of accounts [Mue19]. With today's language models, the ratio of human operators to accounts would be far more extreme. A single operator with API access could plausibly supervise tens of thousands of accounts, each producing original, persona-consistent content around the clock.

7.6 Closing the Loop: Detection of Simulator-Generated Bots

The simulator includes the BiLSTM detector from the first experiment as an integrated component. After any bot has accumulated posts, its account metadata and post history can be passed to the classifier, which returns a bot probability score. This creates a feedback loop: the same interface used to generate bots can be used to evaluate how detectable they are.

In informal testing during development, the classifier assigned high bot probabilities to most generated accounts, primarily because their metadata statistics, such as the ratio of followers to following, the account age relative to post count, and the regularity of posting intervals, do not follow the distributions present in the Cresci-2017 training data. This is an honest limitation: the classifier was trained on a dataset that predates modern generative AI, and the statistical fingerprints of AI-generated content are genuinely different from those of the 2017-era bots it learned from. Re-training on more recent data, including accounts generated by tools like the ones in this simulator, would likely produce a substantially different result. This points to an important conclusion: bot detection systems degrade over time not because the underlying approach is wrong, but because the bots they are trying to detect are themselves evolving. The classifier and the generator are locked in an adversarial co-evolutionary process, and the generator is currently moving faster.

7.7 Summary

This experiment brought together four generative systems, a large language model for text, a latent diffusion model for photographs, a video diffusion model for clips, and the BiLSTM classifier from the first experiment for detection, into a single interactive environment. Each model contributes a layer of authenticity that alone would be unremarkable but in combination produces something genuinely difficult to dismiss at a glance.

The broader point is architectural. Modern AI capabilities have lowered every barrier that previously made running a convincing bot farm expensive: writing quality, photographic identity, video production, and conversational depth. What remains hard is not producing the content but producing it at the scale and coordination that distinguishes an influence operation from a few convincing fake accounts. The statistical signals that the first experiment's classifier exploits, posting regularity, metadata ratios, and stylistic consistency across large volumes of posts, remain informative precisely because they reflect operational constraints that gen-

Figure 7.4: The detection view. The BiLSTM classifier produces a confidence score based on account metadata and post history.

erative models have not yet eliminated. They may not stay informative forever.

Chapter 8

Counter-Strategies and Platform Defenses

The experimental chapters of this dissertation address detection: the technical problem of identifying automated accounts. Detection, however, is only one component of a broader defensive ecosystem. This chapter steps back to consider that ecosystem as a whole, examining the defenses available at the platform level, the regulatory level, and the level of the individual user. It pays particular attention to the gap, made vivid by the Romanian case, between what is technically possible and what is actually deployed.

8.1 Platform-Level Defenses

Social media platforms occupy a privileged position in the defensive landscape: they alone possess complete visibility into account behavior, network structure, and content. They deploy a range of defensive measures, though the details are proprietary and rarely disclosed.

The most basic defenses operate at the point of account creation and action. Rate limiting restricts how frequently an account can post, follow, or message, raising the cost of high-volume automation. CAPTCHA challenges attempt to distinguish human from automated actors at sensitive moments. These measures raise the barrier to entry but are routinely circumvented by sophisticated operations, which distribute activity across many accounts to stay below rate thresholds and which can solve or bypass CAPTCHA challenges through commercial services.

More sophisticated platform defenses apply machine learning at scale to detect coordinated inauthentic behavior, the same problem addressed by the academic methods of Chapter 5, but with the advantage of complete data. Platforms periodically announce the removal of large coordinated networks, which demonstrates that these systems function. The difficulty, illustrated repeatedly in the case studies, is one of speed and incentive: detection that occurs after an election has concluded provides little remedy, and platforms face structural incentives that do not always align with aggressive suppression of engagement, even inauthentic engagement.

8.2 The Limits of Automated Moderation

The Romanian and Slovak cases together expose the limits of automated moderation as currently practiced. In Slovakia, platform responses to the deepfake audio were inconsistent across and even within platforms, with some copies removed, others labelled, and others left in place [CNN24]. In Romania, the coordinated operation achieved its effect before any decisive platform intervention, and the most consequential amplification came not from the automated accounts directly but from the recommendation algorithm responding to the engagement they generated [Fun25, Glo24].

This last point deserves emphasis because it complicates the entire defensive framing. If the harm arises not only from automated accounts but from the interaction between those accounts and a recommendation system optimized for engagement, then detection and removal of the accounts addresses only part of the problem. The recommendation algorithm itself becomes part of the attack surface, and defending against manipulation requires scrutiny of the amplification machinery, not only of the accounts that exploit it.

8.3 Regulatory Responses

Regulation has emerged as a parallel track to technical defense, motivated by the recognition that platforms cannot be relied upon to police themselves adequately where commercial incentives conflict with the public interest. The European Union's Digital Services Act, which entered full force in 2024, is the most significant instrument to date. It imposes on very large online platforms a set of obligations to assess and mitigate systemic risks, including risks to electoral integrity and risks arising from disinformation [Eur24].

The Romanian case became the first major test of this framework, with the European Commission opening formal proceedings against TikTok concerning its handling of the election [Eur24]. The proceedings illustrate both the promise and the limits of the regulatory approach. On one hand, the Digital Services Act provides, for the first time, a legal mechanism to compel platform accountability and to demand transparency about systemic risks. On the other hand, regulatory processes operate on timescales of months and years, while an influence operation achieves its effect in days, and the evidentiary and attributional challenges of demonstrating manipulation through opaque algorithmic systems remain formidable. Researchers have documented the practical difficulty of obtaining from platforms even the data needed to study these phenomena, a tension that the Digital Services Act's transparency provisions are intended, but not yet proven, to resolve [R+25].

8.4 User-Facing Tools and Media Literacy

A third line of defense operates at the level of the individual user. Tools such as the Botometer API allow users and researchers to assess the likelihood that a given account is automated, and browser extensions and similar utilities aim to surface this information at the point of consumption. The reach of such tools is inherently limited, however, since they are used predominantly by the already-skeptical minority who seek them out.

Media literacy education represents the more fundamental, if slower, intervention. A population equipped to recognize the hallmarks of coordinated manipulation, to be appropriately skeptical of content that arrives through algorithmic amplification, and to verify claims before sharing them is more resilient to influence operations than any technical filter. The limitation is one of timescale and reach: media literacy is a generational project, and influence operations exploit precisely the cognitive shortcuts that education seeks to correct.

8.5 Lessons from Romania

The Romanian case, as the most consequential instance examined in this dissertation, offers several concrete lessons for the design of defenses. First, the diversity of the account infrastructure, with distinct IP addresses per account, defeated the simplest network-level defenses, indicating that effective detection must rely on behavioral and relational signals rather than on shared-infrastructure heuristics. Second, the activation of long-dormant accounts defeated recency-based risk scoring, indicating that account age is not a reliable proxy for legitimacy. Third, and most importantly, the central role of algorithmic amplification indicates that defending electoral integrity requires attention to recommendation systems as well as to the accounts that exploit them.

Most broadly, the Romanian case reveals a coordination failure across the defensive ecosystem. Multiple agencies received indications of coordinated activity, but no single actor held both the authority and the timely information needed to respond before the damage was done [BIS25]. This is not primarily a technical failure; it is a failure of institutional design, and it suggests that the most important improvements may lie less in better detection algorithms than in clearer lines of responsibility and faster channels between technical detection and institutional response.

8.6 Synthesis

No single defensive layer is sufficient. Platform defenses possess the data but lack the incentive and the speed; regulation possesses the authority but lacks the timescale; user-facing tools and media literacy possess legitimacy but lack reach. A credible defense against modern influence operations requires all three layers working in coordination, informed by the kind of technical detection capability that the experimental work of this dissertation seeks to advance. Detection, in other words, is necessary but not sufficient: it is the sensing component of a defensive system whose effectiveness ultimately depends on how quickly and coherently the rest of the system can act on what the sensors report.

Chapter 9

Conclusions

This dissertation set out to look at social media bots at a moment when generative AI is changing what they are. The theoretical chapters traced how detection has moved through three generations, with the third one, the one we are living through now, defined by language models and diffusion images that quietly dismantle the features earlier systems relied on. The Romanian presidential election of 2024 sits at the centre of that story as the first European vote annulled because of bot-driven interference, and it kept the technical discussion tied to something real.

The two experiments approached the problem from opposite ends. The first built a Bidirectional LSTM detector that combined account metadata with tweet embeddings and reached 99.2 percent on Cresci-2017. The number looks good and it is also slightly misleading, for reasons the chapter laid out openly: the dataset has known sampling biases, and a high score on it is not the same as a system that would work in deployment. The second experiment built a small generative bot farm with Llama 3.1, Stable Diffusion, and LTX-Video, and ran the detector against it. The detector caught most of the synthetic accounts, but mostly because their metadata patterns did not match anything from 2017. A model trained on data from before modern generative AI catches modern bots by accident more than by design, and that, more than the accuracy figure, is the finding that matters.

The thread running through both experiments is that the cost of producing a convincing fake identity has collapsed, while the cost of catching one has not fallen at anywhere near the same speed. The Romanian case shows what that asymmetry looks like when it reaches an election. Closing the gap will need fresher datasets, cross-dataset evaluation as a default rather than an afterthought, and detection methods that lean on the behavioural and relational signals that are still hard to fake at scale. None of that is solved here. What this dissertation has tried to do, more modestly, is describe the problem clearly enough that the next attempt to solve it has somewhere honest to start.

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